

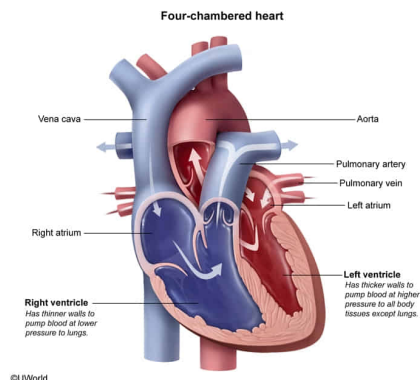
Compared to the blood flowing into the left ventricle, the blood flowing into the right ventricle:

- ☐ A. has a higher oxygen concentration. [10%]
- ☐ B. has a higher pH. [18%]
- ☒ C. is entering a cardiac chamber with thinner walls. [58%]
- ☐ D. is entering a cardiac chamber that contracts to produce higher pressure. [12%]

Correct

58% answered correctly

Explanation:



The **heart** is a muscular structure with **four chambers**: The **left and right atria** receive blood and the **left and right ventricles** then pump this blood through the circulatory system. The heart is composed of two separate pumps that contract simultaneously; the right heart pumps blood to the lungs, whereas the left heart pumps blood to the rest of the body.

Because the left ventricle must pump blood to a *larger* structure (all body tissues except the lungs) across *longer* distances, it must pump blood at a *higher* pressure. To generate the high pressure necessary to propel blood through the body's entire circulatory system, the walls of the left ventricle must be extremely muscular and powerful (ie, thicker). In contrast, the **right ventricle** has **thinner walls than the left ventricle** because it needs to pump blood at a *lower (not higher)* pressure to propel blood to the lungs only (**Choice D**).

(Choice A) **Blood flowing** through the right ventricle has *lower* oxygen concentration than blood flowing through the left ventricle. This is because *deoxygenated blood* from body tissues passes through the *right heart* (right atrium and right ventricle) before being sent to the lungs. *Oxygenated blood* from the lungs passes through the *left heart* (left atrium and left ventricle) before being sent to the body.

(Choice B) The following reaction takes place in **red blood cells**:



The higher the CO_2 concentration, the more the forward reaction is favored. Therefore, the deoxygenated (CO_2 -rich) blood in the right ventricle also contains *more* H^+ ions and so has a slightly *lower* **pH** than the oxygenated (CO_2 -poor) blood of the left ventricle.

Educational objective:

The right ventricle has thinner walls than the left ventricle because it needs to pump blood at a lower pressure to reach only the lungs. In contrast, the thicker walls of the left ventricle allow it to pump blood at a higher pressure to reach all other tissues in the body.

Time spent: 0 sec

Subject:
Biology

QID: 401613

System:
Circulation and Respiration